Shanilka Haturusinghe

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Education

University of Kelaniya, Sri Lanka. March 2020 – October 2024
Bachelor of Science Honours in Computer Science *(First Class)*
Specialization: Artificial Intelligence
`GPA: 4.0/4.0`
Graduated ranked 1st in the Faculty of Computing and Technology (Class of 2025), across all degree programs.

Royal College, Colombo, Sri Lanka.
G.C.E Advanced Level in Mathematics Stream (English Medium)
Results: A in Combined Mathematics, B in Physics, B in Chemistry

Publications

Shanilka Haturusinghe, Tharindu Cyril Weerasooriya, Christopher M Homan, Marcos Zampieri, and Sidath Ravindra Liyanage. 2025. *Subasa – Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala.* In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 4: Student Research Workshop)*, pages 260–270, Albuquerque, USA. Association for Computational Linguistics. DOI: 10.18653/v1/2025.naacl-srw.26
https://aclanthology.org/2025.naacl-srw.26/

Experience - Acedemic

University of Kelaniya February 2025 - Present
Teaching Assistant & Demonstrator
- Serve as Teaching Assistant for the courses *DSCI 32012 - Advanced Database Applications*, *CTEC 22043 - Object Oriented Programming*, *AINT 32012 - Digital Image Processing and Computer Vision*, and *CSCI 22082 - Object-Oriented Programming — II*.
- Conduct tutorials and lab sessions for the aforementioned courses.
- For *CSCI 22082*, led the practical-based lectures for Spring Boot: planned and conducted comprehensive, project-based lessons that progressively covered the fundamentals up to advanced topics in MVC application development in Spring Boot.
- Assisted faculty members in updating the course syllabus for better alignment with industry standards.
- **`Key Technologies: Java, Spring Boot, Python, OpenCV, SQL, PL/SQL Oracle Database XE 18`**

Experience - Industry

Embla Software Innovation (Pvt) Ltd October 2023 - April 2024
Software Engineer Intern
- Developed and implemented the "Scrapbook" feature for Embla FT genealogy application with customizable page templates, editable text/photo elements, and comprehensive image/content management capabilities.
- Built full-stack components using C#/.NET Core backend and Angular 9/TypeScript frontend, enhancing existing application architecture to support new scrapbook functionality and user interactions.
- Collaborated with product teams using Agile methodologies, Jira, and Bitbucket for project delivery, utilizing Visual Studio, FastReport Designer, and Figma for development and design tasks.
- **`Key Technologies: C#, Angular 9, .NET Core, TypeScript, JavaScript, REST Web API, FastReport, Figma`**

Xempler Pte. Ltd September 2022 - May 2023
Intern - Software Developer
- Developed a Sri Lanka License Plate Detection proof of concept using Python and TensorFlow by leveraging YOLO models for robust license plate detection in images. Employed OpenCV for image preprocessing (resizing, filtering, and edge detection) to optimize detection accuracy, and integrated Tesseract OCR to extract alphanumeric text from the detected license plates.
- Designed and implemented a real-time chat feature for the IFAWPCA mobile app, built with React Native. Integrated Firebase for seamless real-time messaging, utilizing Firebase Realtime Database and Firebase Authentication for user management, secure login, and instant message synchronization across devices.
- **`Key Technologies: Angular, Laravel, Python, TensorFlow, OpenCV, Tesseract OCR, React Native, Firebase, Git`**

Experience - Research

"Subasa" Research Project with Advisors: Dr. S.R. Liyanage & Dr. Tharindu Cyril Weerasooriya
October 2023 – December 2024
Conducted the *Subasa* project as first author, developing and evaluating low-resource offensive language detection models for Sinhala using novel strategies.

- Developed *Subasa-XLM-R* using a two-stage intermediate pre-finetuning strategy with Masked Rationale Prediction (MRP), where rationale embeddings are fused with token embeddings and 75% of non-special tokens are masked to optimize learning signal for Sinhala context.
- Implemented task-specific instruction fine-tuning for Llama-3 and Mistral models using PEFT with QLoRA, incorporating classification and offensive phrase extraction in a unified prompt framework.
- Conducted comprehensive ablation studies on masking ratios , intermediate task replacement , and direct fine-tuning approaches to validate the effectiveness of the intermediate pre-finetuning strategy.
- Achieved state-of-the-art performance on SOLD benchmark dataset with Subasa-XLM-R (Macro F1: 0.84), outperforming GPT-4o zero-shot performance and existing baselines.
- Open-sourced complete model suite and codebase.
- `Key Technologies: Python, PyTorch, Hugging Face Transformers, LoRA/QLoRA, Google Colab, Google Cloud Compute Engine`

Personal Projects

PersonaCraft.AI - Personalized AI Writing Assistant https://github.com/haturusinghe/persona-craft-ai

- Developed PersonaCraft.AI, an end-to-end ML system for creating personalized AI writing assistants through automated data collection, processing, and model fine-tuning with multi-platform web scraping capabilities.
- Built automated data extraction pipelines using Selenium for LinkedIn, Medium, GitHub, and custom web sources, implementing advanced text processing with chunking and embedding generation via Sentence Transformers.
- Engineered comprehensive ML training pipeline supporting both Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) using TRL and Unsloth,.
- Implemented MLOps architecture using ZenML for pipeline orchestration, MongoDB for data warehousing, Qdrant vector database for RAG-based retrieval, and AWS SageMaker for cloud-based model training.
- Integrated automated publishing workflow with Hugging Face Hub for dataset and model sharing, including model validation and monitoring capabilities.
- `Key Technologies: Python, ZenML, LangChain, GPT-4, Llama 3.1, TRL, Unsloth, MongoDB, Qdrant, AWS SageMaker, Selenium, Docker, Poetry, Comet ML`

Real-Time Fashion Recommendation System https://github.com/haturusinghe/realtime-fashion-recommender

- Developed a MLOps-driven fashion recommendation system using H&M's Kaggle dataset, implementing a 4-stage architecture with decoupled ML pipelines for feature processing, training, and inference.
- Implemented a two-tower neural network architecture for customer&item embedding generation, utilizing Hopsworks Vector Index for embedding storage and retrieval.
- Built comprehensive ML pipelines including feature engineering, retrieval model training (two-tower architecture), ranking model training (CatBoost), item embedding computation, and model deployment with real-time inference capabilities.
- Integrated Hopsworks Feature Store for feature management, Model Registry for versioning, and Model Serving for deployment, with an interactive Streamlit frontend for customer recommendations and feedback.
- `Key Technologies: Python, TensorFlow, Hopsworks, Sentence-Transformers, CatBoost, Polars, Streamlit`

Deep Learning based real-time Facial Age and Gender Detection System

https://github.com/haturusinghe/cnn-age-gender-detection-android

- Developed CNN models for age and gender detection, integrated into Android and web applications for real-time predictions. Backend implemented with TF-Serving and Flask, and frontend built using React.
- Deployed system using TensorFlow for model inference and integrated a model-serving pipeline with preprocessing. Ensured real-time interaction through web and mobile interfaces.
- `Key Technologies: Python, Flask, TensorFlow (+ TF-Lite, TF-Serving), React, Android Studio, Kotlin, Heroku`

Awards

- **Gold Medal** for Academic Excellence (2025) – Awarded by the Alumni Association of the University of Kelaniya for achieving the highest GPA (4.0/4.0) in the Faculty of Computing and Technology, with First Class Honours.
- **Gold Medal** for Outstanding Performance (2025) – Awarded by the Faculty of Computing and Technology for attaining the highest GPA (4.0/4.0) in the B.Sc. Honours in Computer Science degree program, with First Class Honours.
- **Google Cloud Research** Credits Program (Award GCP19980904) – Awarded **$1000** in research credits

Extra Curricular

Huawei Seeds for the Future - 2023

- Team Leader for Team Sri Lanka at Huawei Seeds for the Future - 2023 – Selected as one of only four top university students from a highly competitive pool, earning the opportunity to visit China.
- Received mentorship from top industry experts on cutting-edge technologies including 5G, AI, Cloud Computing, and Digital Power.
- Participated in the Global Tech4Good competition, honing skills in leadership, problem-solving, and entrepreneurship.

References

Prof. Sidath R. Liyanage
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